

INSPECT

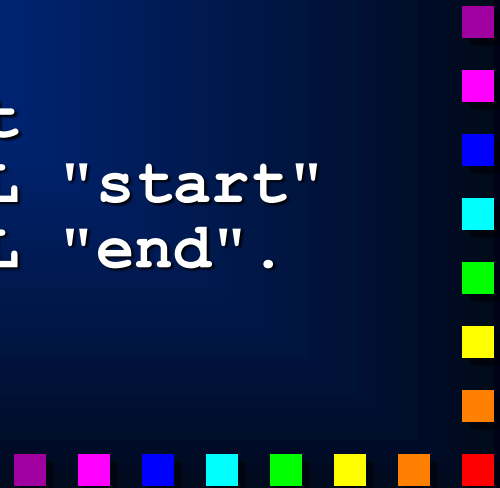
INSPECT Syntax - Format 1

INSPECT SourceStr\$i TALLYING

Counter#i FOR $\left\{ \begin{array}{l} \underline{\text{CHARACTERS}} \left[\begin{array}{l} \underline{\text{BEFORE}} \\ \underline{\text{AFTER}} \end{array} \right] \text{INITIAL Delim$il} \dots \\ \left\{ \begin{array}{l} \underline{\text{ALL}} \\ \underline{\text{LEADING}} \end{array} \right\} \left\{ \text{Compare$il} \left[\begin{array}{l} \underline{\text{BEFORE}} \\ \underline{\text{AFTER}} \end{array} \right] \text{INITIAL Delim$il} \dots \right\} \dots \end{array} \right\} \dots$

INSPECT FullName TALLYING UnstrPtr
FOR LEADING SPACES.

INSPECT SourceLine TALLYING ECount
FOR ALL "e" AFTER INITIAL "start"
BEFORE INITIAL "end".



INSPECT Example 1

```
READ TextFile
  AT END SET EndOfFile TO TRUE
END-READ
PERFORM UNTIL EndOfFile
  PERFORM VARYING idx FROM 1 BY 1 UNTIL idx > 26
    INSPECT TextLine TALLYING LetterCount(idx)
      FOR ALL Letter(idx)

  END-PERFORM
  READ TextFile
    AT END SET EndOfFile TO TRUE
  END-READ
END-PERFORM
PERFORM VARYING idx FROM 1 BY 1 UNTIL idx > 26
  DISPLAY "Letter " Letter(idx)
    " occurs " LetterCount(idx) " times"
END-PERFORM
```

How the INSPECT works.

The INSPECT scans the **Source String** from left to right **counting** and/or **replacing** characters under the control of the **TALLYING**, **REPLACING** or **CONVERTING** phrases.

The **behaviour** of the INSPECT is **modified by** using the **LEADING**, **FIRST**, **BEFORE** and **AFTER** phrases.

An **ALL**, **LEADING**, **CHARACTERS**, **FIRST** or **CONVERTING** phrase may only be followed by **one BEFORE** and **one AFTER** phrase.



Modifying Phrases

LEADING

The LEADING phrase causes counting/replacement of all Compare\$il characters from the **first valid one** encountered to the **first invalid one**.

FIRST

The FIRST phrase causes only the **first valid character** to be replaced.

BEFORE

The BEFORE phrase designates as valid those characters to the **left of the delimiter** associated with it.

AFTER

The AFTER phrase designates as valid those characters to the **right of the delimiter** associated with it.



INSPECT Syntax - Format 2

INSPECT SourceStr\$ REPLACING

$$\left\{ \begin{array}{l} \text{CHARACTERS BY ReplaceChar\$il} \left[\left\{ \begin{array}{c} \text{BEFORE} \\ \text{AFTER} \end{array} \right\} \text{INITIAL Delim\$il} \right] \dots \\ \left\{ \begin{array}{c} \text{ALL} \\ \text{LEADING} \\ \text{FIRST} \end{array} \right\} \left\{ \text{Compare\$il BY Replace\$il} \left[\left\{ \begin{array}{c} \text{BEFORE} \\ \text{AFTER} \end{array} \right\} \text{INITIAL Delim\$il} \right] \dots \right\} \dots \end{array} \right\} \dots$$

PERFORM VARYING idx FROM 1 BY 1 UNTIL idx > 10

INSPECT TextLine

REPLACING SwearWord(idx) BY "*#@!"

END-PERFORM



INSPECT Example 2

INSPECT StringData REPLACING ALL "F" BY "G"
AFTER INITIAL "A" BEFORE INITIAL "Q".

StringData

F	F	F	F	A	F	F	F	F	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 2

INSPECT StringData REPLACING ALL "F" BY "G"
AFTER INITIAL "A" BEFORE INITIAL "Q".

StringData

F	F	F	F	A	G	G	G	G	G	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 3

INSPECT StringData REPLACING LEADING "F" BY "G"
AFTER INITIAL "A" BEFORE INITIAL "Z".

StringData

F	F	F	F	A	F	F	F	F	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 3

INSPECT StringData REPLACING LEADING "F" BY "G"
AFTER INITIAL "A" BEFORE INITIAL "Z".

StringData

F	F	F	F	A	G	G	G	G	G	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 4

INSPECT StringData REPLACING ALL "F" BY "G"
AFTER INITIAL "A" BEFORE INITIAL "Z".

StringData

F	F	F	F	A	F	F	F	F	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 4

INSPECT StringData REPLACING ALL "F" BY "G"
AFTER INITIAL "A" BEFORE INITIAL "Z".

StringData

F	F	F	F	A	G	G	G	G	G	Q	G	G	G	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 5

INSPECT StringData REPLACING FIRST "F" BY "G"
AFTER INITIAL "A" BEFORE INITIAL "Q".

StringData

F	F	F	F	A	F	F	F	F	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 5

INSPECT StringData REPLACING FIRST "F" BY "G"
AFTER INITIAL "A" BEFORE INITIAL "Q".

StringData

F	F	F	F	A	G	F	F	F	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 6

INSPECT StringData REPLACING
ALL "FFFF" BY "FROG"
AFTER INITIAL "A" BEFORE INITIAL "Q".

StringData

F	F	F	F	A	F	F	F	F	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 6

INSPECT StringData REPLACING
ALL "FFFF" BY "FROG"
AFTER INITIAL "A" BEFORE INITIAL "Q".

StringData

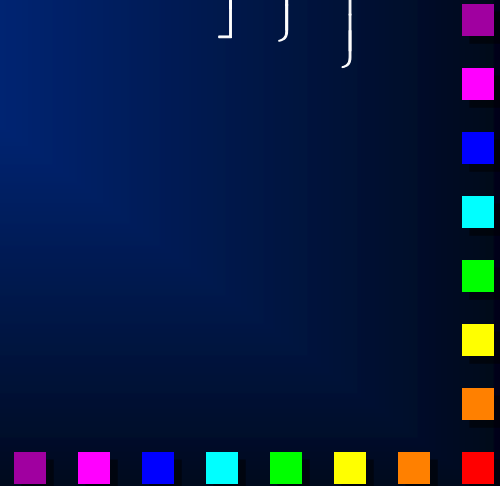
F	F	F	F	A	F	R	O	G	F	Q	F	F	F	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Syntax - Format 3

INSPECT SourceStr\$ TALLYING

Counter# i FOR { CHARACTERS [{ BEFORE } INITIAL Delim\$il] ... }
 { { ALL } { De lim \$il [{ BEFORE } INITIAL Delim\$il] ... } ... } ...
 { LEADING }

REPLACING { CHARACTERS BY ReplaceChar\$il [{ BEFORE } INITIAL Delim\$il] ... }
 { { ALL } { Compare\$il BY Replace\$il [{ BEFORE } INITIAL Delim\$il] ... } ... } ...
 { LEADING }
 { FIRST }



INSPECT Syntax - Format 4

INSPECT SourceStr\$i CONVERTING

Compare\$il TO Convert\$il

$\left[\left\{ \begin{array}{c} \underline{\text{BEFORE}} \\ \underline{\text{AFTER}} \end{array} \right\} \text{INITIAL Delim$il} \right] \dots$

INSPECT TextLine CONVERTING

"0123456789" TO "5298317046"

AFTER INITIAL "codeon"

BEFORE INITIAL "codeoff".



INSPECT Example 7

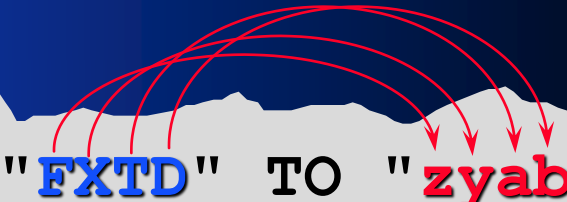
INSPECT StringData CONVERTING "FXTD" TO "zyab".

StringData

P	X	P	F	P	D	P	T	P	P	F	X	T	D	P
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 7

INSPECT StringData CONVERTING "FXTD" TO "zyab".

A diagram showing the conversion of the string "FXTD" to "zyab". Red curved arrows connect the characters: 'F' to 'z', 'X' to 'y', 'T' to 'a', and 'D' to 'b'.

StringData

P	y	P	z	P	b	P	a	P	P	z	y	a	b	P
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 7

INSPECT StringData REPLACING "FXTD" BY "zyab".

StringData

P	X	P	F	P	D	P	T	P	P	F	X	T	D	P
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 7

INSPECT StringData REPLACING "FXTD" BY "zyab".

StringData

P	X	P	F	P	D	P	T	P	P	z	y	a	b	P
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 8

INSPECT StringData REPLACING ALL "x" BY "y"
"d" BY "z"
"f" BY "s".

StringData

P	x	P	f	P	d	P	T	P	P	f	x	T	d	P
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 8

INSPECT StringData REPLACING ALL "X" BY "y"
"D" BY "z"
"F" BY "s".

StringData

P	y	P	s	P	z	P	T	P	P	s	y	T	z	P
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

INSPECT Example 9

```
INSPECT CustAddress  
  CONVERTING "abcdefghijklmnopqrstuvwxyz"  
    TO      "ABCDEFGHIJKLMNOPQRSTUVWXYZ".
```

```
01 AlphaChars.  
  02 AlphaLower PIC X(26) VALUE  
    "abcdefghijklmnopqrstuvwxyz".  
  02 AlphaUpper PIC X(26) VALUE  
    "ABCDEFGHIJKLMNOPQRSTUVWXYZ".
```

```
INSPECT CustAddress  
  CONVERTING AlphaLower TO AlphaUpper.  
INSPECT CustAddress  
  CONVERTING AlphaUpper TO AlphaLower.
```

INSPECT Example 10

```
01 RussianPay PIC $$$,$$$,$$9.99.
```

```
MOVE 12345.67 TO RussianPay.  
INSPECT RussianPay REPLACING ALL "$" BY "R".  
DISPLAY RussianPay " roubles".
```

RussianPay

INSPECT Example 10

```
01 RussianPay PIC $$$,$$$,$$9.99.
```

```
MOVE 12345.67 TO RussianPay.
```

```
INSPECT RussianPay REPLACING ALL "$" BY "R".  
DISPLAY RussianPay " roubles".
```

RussianPay

\$12,345.67

INSPECT Example 10

```
01 RussianPay PIC $$$,$$$,$$9.99.
```

```
MOVE 12345.67 TO RussianPay.
```

```
INSPECT RussianPay REPLACING ALL "$" BY "R".
```

```
DISPLAY RussianPay " roubles".
```

RussianPay

R12,345.67

INSPECT Example 10

```
01 RussianPay PIC $$$,$$$,$$9.99.
```

```
MOVE 12345.67 TO RussianPay.  
INSPECT RussianPay REPLACING ALL "$" BY "R".  
DISPLAY RussianPay " roubles".
```

RussianPay

R12,345.67

R12,345.67 roubles